How Much Personal History Is Enough? Measuring the Data Requirement for Individualised Financial Behavioural Anomaly Detection

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***Abstract*—A personalised anomaly detector compares each transaction with a model of one person's past, but a new customer has none, and how much history is enough has not been measured. Measuring it is hard because real data lack anomaly labels, because ranking quality does not provide an alert threshold, and because a per-person score built from a few anomalies is noisy. We hold each person's evaluation window fixed while the history grows, label anomalies by provenance (a transaction is replaced by another person's real transaction, at a 10% rate; these are not fraud), set thresholds by three rules including a conformal one, and redraw each injection 20 times to separate individual differences from noise. On 1990s accounts from one Czech bank, a Mahalanobis detector reaches 90% of its best precision-recall AUC (about 0.66) after 10 to 22 transactions, depending mainly on whether calendar features are used. A population model is better below about ten transactions; shrinking toward it removes that penalty, but afterwards the personal model is at least as good. A conformal threshold needs 19 transactions and is exceeded on long histories. People differ genuinely in how well they can be modelled, but observed behavioural variability does not predict how much history they need. On retail purchase lines a single amount feature saturates after one line, and on synthetic card data the best detector is still improving at 500 transactions. The sufficiency point depends on detector, features, anomaly definition and data, and must be measured per system.**

***Index Terms*—anomaly detection, personalisation, cold start, learning curves, transaction data, conformal calibration, measurement reliability**

I. Introduction

A personalised anomaly detector compares each new transaction with a model of one person's own past. A 900-unit purchase is routine for one account holder and extraordinary for another, and a single population-wide rule serves both badly. The personal model, however, has to be estimated from the person's own transactions, and a new account has none. Somewhere between zero and a long history the personal baseline becomes worth using. Fraud researchers have long built per-account profiles and population baselines side by side [1]–[3], and behaviour-monitoring systems simply exclude users "with insufficient history" [4]. How much history is sufficient, and what "sufficient" should mean, has not been measured.

Measuring it is harder than sweeping training-set size. Three problems stand in the way. First, real transaction data rarely carry anomaly labels, and labelling anomalies with a statistical rule and then evaluating detectors that compute something similar is circular. Second, a detector that ranks anomalies well is not yet usable: it also needs an alert threshold, which must be set from the person's own short history and whose error rate is easy to misjudge. Third, a per-person evaluation rests on a handful of anomalies, so the per-person score is a noisy measurement, and noise can masquerade as individual differences.

We build a measurement protocol that addresses each problem and apply it to two real datasets and two synthetic ones. Each entity's evaluation window is held fixed while the personal history grows, so every comparison across history sizes is paired. Ground truth is provenance: a window transaction is replaced by a real transaction of a different entity, the design used to evaluate per-user detectors in keystroke dynamics [5]. Alert thresholds are set by three rules, including a conformal rule with a finite-sample guarantee [6], and each is compared with its own exact reference. Every entity's injection is redrawn 20 times, which separates true individual differences from measurement noise.

On 300 real bank accounts at each of five cohort draws, a ridge-regularised Mahalanobis detector reaches 90% of its best ranking quality after 10 to 22 of the person's own transactions: about 20 when calendar features are included and anomalies are defined by impostor substitution, 10 to 12 without calendar features. The number depends on the features, the anomaly definition, the metric and the dataset, and the paper is as much about that dependence as about the number.

Contributions:

1) **A measured sufficiency point and what it depends on.** Paired, provenance-labelled data-efficiency curves for eight unsupervised detectors. On bank data the relative-90 threshold for Mahalanobis is 20, 20, 22, 20, 22 across cohort draws with calendar features under impostor substitution, and 10 to 12 without calendar features, whose contribution to the ceiling appears only under that protocol. On retail purchase lines, which arrive in invoices, a one-feature amount score saturates after a single line and the multivariate detectors barely learn; on synthetic card data the best detector is still improving at 500 transactions (Sections V-A, V-B and V-F).
2) **A crossover and a narrower remedy.** On bank data a population model outperforms the personal one below $n = 10$ at every cohort draw. Shrinking the personal estimate toward the population removes that penalty under every anomaly definition, but after the crossover the pre-specified hybrid trails the personal model under

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most definitions and metrics, and where the population model is weak, as on retail data, it costs accuracy from the start (Sections V-C and V-F).

3) **Ranking quality is not an operating point.** In-sample thresholds flag most genuine transactions at small n; quantile thresholds must be judged against their finite-sample reference, not the nominal rate; a conformal threshold needs 19 calibration points, holds its 5% guarantee from 20 to 50 transactions on bank data, and is exceeded on long bank histories, on retail data and in simulation, in each case consistent with a past that is not exchangeable with the present (Sections V-D and V-F).
4) **Individual differences, measured properly.** On bank data a single-draw per-person score at $n = 25$ is more noise than signal (intraclass correlation 0.40), yet true differences between people remain (standard deviation 0.119 at $n = 25$). Observed behavioural variability predicts how well a person can be modelled, but not how much history they need. In simulation, behavioural regimes set by design change the requirement roughly tenfold (Section V-E).

Section II positions the work, Sections III and IV define the data and protocol, Section V reports results, and Section VI draws out what the evidence establishes, its limitations and its ethical implications.

II. Related Work

a) Account profiling in fraud detection: Comparing an account with its own past is one of the oldest ideas in statistical fraud detection. Bolton and Hand review it alongside supervised scoring [1]. Their peer-group analysis replaces the single account's history with a group of accounts that behaved similarly in the past, explicitly because an account's own record is short and noisy [2]. Whitrow *et al.* show that how much past activity is aggregated into each transaction's features changes detection performance substantially [7], and Juszczak *et al.* compare generic and account-tailored one-class and two-class classifiers on two large personal-banking datasets [3]. The same profiling logic appears in telecommunications fraud [8]. Later card-fraud work builds per-cardholder aggregates [9], models a card's transaction stream as a sequence [10], combines unsupervised scores with supervised ones [11] and studies realistic evaluation under alert feedback [12]–[14]. This literature establishes that personal baselines and population baselines both carry signal. It reports performance at the history lengths available in each dataset; it does not measure performance as a function of the number of the individual's own transactions under a design that holds the evaluation set fixed. Deployed behaviour-profiling systems treat that number as a configuration parameter; two patent families in user-activity security monitoring, for example, build per-user baselines and exclude users "with insufficient history" [4], [15].

b) Per-user anomaly detection outside finance: Keystroke-dynamics research faced the same design question earlier and answered it with the evaluation design we adopt. Killourhy and Maxion benchmark per-user anomaly detectors by training on one typist's samples and scoring other typists' samples as impostors [5], and their follow-up finds that the amount of training data is among the most influential factors in per-user error and that some users are much easier to separate than others [16]. We transfer this impostor protocol to transaction histories and ask the question they treated as one factor among several as the primary question.

c) Unsupervised detectors: We use the standard families surveyed in [17]–[19]: univariate deviation scores, Mahalanobis distance, nearest-neighbour distance and local outlier factor [20], isolation forests [21], [22] and a reconstruction-error autoencoder. Benchmarks find that no family dominates across datasets [23], [24], which is one reason we report every detector at every history size rather than a single winner. Because personal histories are short relative to the feature dimension, covariance estimation is the crux for distance-based detectors; we compare a ridge-regularised estimate with Ledoit–Wolf shrinkage [25], [26].

d) Cold start and shrinkage toward a population: Recommender systems call the thin-history problem cold start and address it by borrowing strength from other users [27], [28], sometimes reporting accuracy across per-user interaction caps [29]; graph methods do the same for fraud by aggregating over related accounts [30]. The statistical template for borrowing strength is shrinkage of individual estimates toward a group estimate with a weight set by the amount of individual data [31]. Our hybrid is this template applied to a mean and covariance: its location update is the posterior mean of a normal model with a conjugate prior worth κ pseudo-observations. We do not propose the estimator; we measure where it helps and where it does not.

e) Learning curves and sample size: Learning curves relate performance to training-set size; their shape is task-dependent and need not be monotone [32], [33], and they are used to plan how much data a model needs [34]. Our curves are per-individual learning curves: the training set is one person's history, and the evaluation set is that person's own later transactions.

f) Threshold calibration: An anomaly detector needs an alert threshold, and setting it at a quantile of the scores of the points the model was fitted to is optimistic, the familiar bias of in-sample error. Conformal prediction gives a distribution-free alternative: under exchangeability, a threshold at a specific order statistic of out-of-sample calibration scores bounds the false-alarm rate [35]. Conformal anomaly detection applies this to novelty detection [36], and conformal p-values for outlier testing come with finite-sample guarantees [6]; jackknife-style variants reuse all training points as calibration points at a bounded cost in coverage [37]. A consequence that matters for small personal histories is that with m calibration scores no threshold can guarantee a false-positive rate of α unless $m \geq 1/\alpha - 1$, which is 19 for $\alpha = 0.05$.

g) Evaluation with few positives: Precision-recall summaries are the appropriate ranking measure at low prevalence [38], [39], but their sampling variance is large when the number of positives is small [40]. In a per-entity design each entity's score rests on a handful of anomalies, so a per-entity score is a noisy measurement. We quantify its test-retest reliability with intraclass correlation [41], [42] and correct correlations

TABLE I
DATA SOURCES. "ELIGIBLE" ENTITIES HAVE AT LEAST $n_{\max} + 60$ TRANSACTIONS.

Source	Kind	Entity	Eligible	$n_{\max}$
Berka (bank)	real, 1993–98	account	1,319	250
Berka, early life	real	account	3,727	50
Online Retail II	real, 2009–11	customer	601	250
TabFormer (card)	synthetic	user–card	4,538	500
Simulation	synthetic	individual	240	500

for attenuation [43], a step the per-user literature above does not take.

h) Gap: To our knowledge, no prior study reports, for unsupervised behavioural anomaly detection on transaction data, performance against the number of the individual's own transactions with the evaluation set held fixed, a matched population and shrinkage comparison at each size, finite-sample-valid operating points, and a separation of individual differences from measurement noise. This statement rests on searches of indexed literature (CrossRef-indexed journals and proceedings, arXiv, patent databases); paywalled full texts were not all available.

III. Problem Setting and Data

A. Setting

An *entity* is one behavioural stream: a bank account, a retail customer or a (user, card) pair. The *personal history size* n is the number of the entity's own transactions available to fit a detector. A detector is unsupervised: it sees the entity's past transactions and never a label. A *behavioural anomaly* is a transaction that deviates from the pattern established by the same entity's earlier transactions. Behavioural anomalies, statistical outliers, suspicious transactions and confirmed fraud are different things: most behavioural anomalies are benign (a holiday, a house move, a new job), and nothing in our labels is fraud.

We ask five questions. How does detection quality grow with n, and where does it saturate (Section V-A)? Does the answer survive changes of anomaly definition, feature set, cohort and dataset (Section V-B)? Below the sufficiency point, how do personal, population and shrinkage-hybrid models compare (Section V-C)? When does a detector that ranks well also have a valid alert threshold (Section V-D)? And how much of the variation between individuals is real, and what explains it (Section V-E)?

B. Datasets

Table I summarises the four data sources. Results from different sources are never pooled.

a) Berka bank transactions: The PKDD'99 Financial dataset [44] holds real, anonymised retail-banking transactions from a Czech bank: 1,056,320 transactions across 4,500 accounts from 1993-01-01 to 1998-12-31, with day-resolution dates, a credit/debit type, an operation mode, an amount, a balance, a characterisation symbol (`k_symbol`) and partner fields. There are no fraud labels. 32.2% of all rows are generated by the bank rather than the customer (interest credited, statement fees and sanction interest), and all of them fall on the last day of a month. Files came from a GitHub mirror because the primary host did not resolve; every table matched the record counts published by the dataset's authors and was byte-identical, after line-ending normalisation, to a second independent mirror. Berka is the only public real dataset we found with per-entity identity, timestamps and deep per-entity histories; the widely used card benchmark lacks a cardholder identifier, and the other candidates are synthetic or describe applications rather than transaction streams.

b) Online Retail II: The UCI Online Retail II data [45], [46] record every purchase line of a UK online gift retailer from 2009-12-01 to 2011-12-09: 797,885 lines with a customer identifier (after removing exact duplicates) from 5,942 customers, each with an invoice timestamp to the minute, a product code, a quantity and a unit price. A transaction here is a purchase line, so n counts lines, and lines on one invoice share a timestamp. This is real behaviour from a different domain with a product channel that Berka lacks; it tests whether the findings are specific to banking.

c) TabFormer: IBM's TabFormer card data [47] are generator-produced: 24,386,900 transactions from 2,000 users, with merchant, merchant category, merchant location and minute-resolution time. We use them to test the design on a schema with merchant, category and geographic channels and with histories long enough for $n = 500$.

d) Controlled simulation: To manipulate behavioural variability directly, we simulate 240 individuals in six archetypes of 40 (regular, moderate, variable, traveller, seasonal, burst). Each archetype fixes a vector of generative parameters, so individuals of one archetype differ in their randomly drawn merchants, categories, locations, typical amount and preferred hour, but in no dispersion parameter: log-amount dispersion, merchant count and Zipf exponent, category count, location count and home-location weight, inter-arrival Gamma shape, hour-of-day concentration and a drift rate. Anomalies come in six classes (amount, merchant, category, geographic, temporal, multivariate), each drawn from a shifted generative distribution; ground truth is the generating branch.

IV. Measurement Protocol

A. Paired Split Design

For each entity the *evaluation window* is a fixed block of 60 transactions (100 in the simulation), and the training history of size n is the n transactions immediately before it:

$$[\,\text{earlier} \mid \underbrace{n\ \text{training}}_{\text{varies}} \mid \underbrace{60\ \text{evaluation}}_{\text{fixed}}\,].$$

No transaction is both trained on and tested on, and because the window is identical across n for a given entity, comparisons across history sizes are paired within entity. We use two cohorts on Berka. In the *mature* cohort the window is an account's last 60 transactions and $n \in \{1, 5, 10, 12, 15, 18, 20, 22, 25, 50, 100, 250\}$; the dense grid between 10 and 25 is where the curve turns. In the *early-life* cohort the window is transactions 51 to 110 of the account's

life and $n \leq 50$, so $n = 50$ means the account's first 50 transactions; this is the cold-start setting itself. Online Retail II uses the mature design with $n \leq 250$ and TabFormer with $n \leq 500$.

The population model is fitted on the earliest 40 transactions of 400 *donor* entities that are disjoint from the evaluated entities. On Berka we evaluate 300 accounts at each of five cohort-selection seeds; the seeds resample 700 accounts from the same 1,319 eligible ones, so cohorts overlap and the seeds measure sensitivity to cohort draw, not independent replication. The early-life cohort uses 1,500 accounts and one seed; Online Retail II and TabFormer use 300 entities at three seeds.

B. Ground Truth by Provenance

Labelling anomalies with a statistical rule and then scoring detectors that compute something similar measures agreement between two versions of one rule. We define ground truth by provenance instead.

a) Protocol I (impostor substitution, primary): A window transaction of the target entity is replaced by a real transaction of a different entity, drawn at random from the donors. The replacement keeps the target's timestamp and takes the impostor's behavioural fields: amount, type, operation and `k_symbol` on Berka; product, quantity and price on Online Retail II; amount, merchant, category, location, channel and error flag on TabFormer. Substitution rather than insertion keeps the inter-arrival structure intact, so timing alone cannot reveal an injected row, but a detector can still notice a mismatch between the impostor's behaviour and the target's calendar slot. Keystroke-dynamics studies evaluate per-user detectors with the same design [5].

b) Protocol IC (customer-initiated, Berka): As Protocol I, but slots and impostor rows are restricted to customer-initiated transactions, excluding the bank-generated interest and fee rows.

c) Protocol P (parametric): A window transaction's amount (unit price on Online Retail II) is multiplied by 5.

Six of the 60 window rows are injected (10% prevalence). For each entity and protocol we draw the injection 20 times with independent seeds. A detector fitted at size n does not depend on the injection, so it scores all 20 draws, and every per-entity metric is the mean over draws.

C. Features

Every feature of a transaction depends only on that transaction, the entity's earlier transactions and, for novelty features, the training history the detector was fitted on. Inter-arrival gaps are measured to the entity's true previous transaction on the full timeline. Berka has 21 features: log amount, a credit indicator, day of month and day of week as sine–cosine pairs, log gap, and one-hot operation (5 levels plus missing) and `k_symbol` (7 plus missing). The account balance is excluded: it is accumulated state, and under Protocol I a substituted row would carry a balance discontinuity that reveals provenance for reasons unrelated to behaviour. Online Retail II has 13 features: log line value, log quantity, log unit price, a return flag, hour and day of week, log gap, log basket size, product popularity among donor lines, and product novelty and frequency relative to the training history. TabFormer has 26: log amount, hour and day of week, log gap, error, chip and online flags, merchant, category and state novelty, a home-state flag and a 13-level category one-hot. Novelty features of training rows are computed against earlier training rows; those of evaluation rows against the whole training history. The population arm omits features that need personal history. Standardisation is fitted on training rows only. The timing features (calendar position, gap and, on Online Retail II, basket size) are dropped in the *no-timing* ablation.

D. Detectors

- **Gaussian** z and **robust** z: $|z|$ of log amount (log line value on Online Retail II) from mean and standard deviation, or median and median absolute deviation.
- **Mahalanobis (ridge)**: squared Mahalanobis distance from the training mean under the sample covariance plus a ridge of $10^{-3} \max(1, \operatorname{tr} \boldsymbol{\Sigma}/d)$.
- **Mahalanobis (Ledoit–Wolf)**: the same distance under the Ledoit–Wolf shrinkage covariance [25].
- k**-NN**: distance to the fifth-nearest training point.
- **LOF** [20]: local outlier factor in novelty mode with $\min(20, n-1)$ neighbours.
- **Isolation Forest** [21]: 200 trees, subsample size $\min(256, n)$ (scikit-learn [48]).
- **Autoencoder**: a d-4-d tanh network trained with full-batch Adam (learning rate 0.01, weight decay 10^{-4}, 300 epochs).

a) Shrinkage hybrid: The hybrid shrinks personal location and scatter toward population estimates with a weight set by n alone:

$$\boldsymbol{\mu}_h = \frac{n\,\boldsymbol{\mu}_p + \kappa\,\boldsymbol{\mu}_g}{n+\kappa}, \qquad \boldsymbol{\Sigma}_h = \frac{n\,\boldsymbol{\Sigma}_p + \kappa\,\boldsymbol{\Sigma}_g}{n+\kappa} + \epsilon\mathbf{I}, \quad (1)$$

where p and g denote personal and population estimates. The location update is the posterior mean of a normal model whose conjugate prior is worth κ observations [31]. The primary value $\kappa = 25$ was fixed before any result was seen; we also report $\kappa \in \{5, 10, 50, 100\}$. Two implementation details matter for interpreting it. The hybrid works in the population model's standardisation, whereas the personal model standardises by the person's own history, so even as $\kappa \to 0$ the hybrid is not identical to the personal model. And the hybrid uses every feature, including the personal novelty features that the population arm omits (none on Berka; merchant, category and state novelty on TabFormer; product novelty on Online Retail II).

b) One transaction: At $n = 1$ no dispersion can be estimated. The z-scores then use unit scale, the Mahalanobis detectors reduce to the ridge term, so they rank by Euclidean distance from the single training transaction, LOF and k-NN use the distance to that transaction, and Isolation Forest gives every transaction the same score, so its PR-AUC equals the prevalence. Results at $n = 1$ describe these fallbacks rather than learned behaviour.

E. Ranking Metric and Sufficiency

The primary metric is precision-recall AUC (PR-AUC, average precision) [38], [39], whose prevalence baseline is 0.10. A *relative-q threshold* is the smallest evaluated n at which the mean PR-AUC curve reaches $q\%$ of its best observed value and stays there; we report $q = 90$ and $q = 95$, and a bootstrap distribution over entities (1,000 resamples). A threshold against the best *observed* value is optimistic when a curve is still rising at the largest n; we flag such curves. Saturating-exponential and power-law fits are compared by AIC.

F. Alert Thresholds

A deployed detector needs a threshold set from training data alone. We compare three rules at nominal level $\alpha = 0.05$. The *in-sample* rule takes the 95th percentile of the scores the fitted model gives its own training points. The *out-of-sample quantile* rule takes the same percentile of out-of-sample training scores: exact leave-one-out refits for the distance-based detectors and the hybrid, and 10-fold cross-fitting (exact leave-one-out for $n \leq 10$) for Isolation Forest, LOF and the autoencoder. The *conformal* rule takes the k-th smallest of the $m = n$ out-of-sample scores with $k = \lceil (m+1)(1-\alpha) \rceil$, and flags nothing when $k > m$ [6], [36].

The reference for each rule matters. If calibration and test scores were exchangeable, the conformal rule would have false-positive rate (FPR) at most α, but only for $m \geq 19$. The percentile rule would have an expected FPR of

$$1 - \frac{(1-g)(j+1) + g(j+2)}{m+1}, \qquad (2)$$

where $h = (m-1)(1-\alpha)$, $j = \lfloor h \rfloor$ and $g = h - j$. This is 0.132, 0.085, 0.068 and 0.054 at $m = 10$, 25, 50 and 250, not 0.05. We compare each rule with its own reference; departures from it measure non-exchangeability between a person's past and present. The population arm is calibrated by split conformal on 2,000 donor rows held out from its fit.

G. Statistics

The entity is the independent unit, so we resample entities. Arm comparisons use paired bootstrap intervals (2,000 resamples) and the Wilcoxon signed-rank test [49] with rank-biserial effect sizes [50], Holm-corrected within each family [51]. Because each entity's score rests on six anomalies per draw, a single-draw per-entity PR-AUC is a noisy measurement [40]. We estimate its intraclass correlation ICC(1) across draws [41], [42], which splits the across-entity variance into true between-entity variance and injection noise, and the split-half (Spearman–Brown) reliability of per-entity sufficiency thresholds. Correlations involving per-entity outcomes are reported with their attenuation-corrected value [43]. Run-level reproduction is enforced by a regression gate: with the original feature construction and a single draw, the pipeline reproduces the separately coded earlier version of the same design to within 3×10^{-16} (Section B).

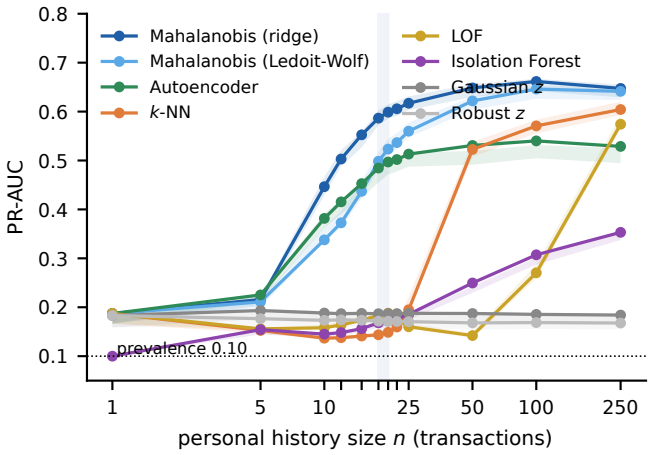


Fig. 1. PR-AUC of personal detectors against personal history size on mature Berka accounts (five cohort seeds, 20 injection draws each). Bands: 95% bootstrap intervals over accounts (primary seed). Shaded column: 95% bootstrap interval of the Mahalanobis relative-90 threshold.

TABLE II
MEAN PR-AUC OF PERSONAL DETECTORS ON MATURE BERKA ACCOUNTS (PROTOCOL I; 300 ACCOUNTS × 5 COHORT SEEDS, 20 INJECTION DRAWS EACH; PREVALENCE 0.100). BEST PER ROW IN BOLD. THE LARGEST ACROSS-SEED STANDARD DEVIATION OF ANY CELL IS 0.015.

n	Mah.	Mah. LW	AE	k-NN	LOF	IF	z
1	0.187	0.187	0.187	0.187	0.187	0.100	0.183
5	0.216	0.211	**0.225**	0.153	0.156	0.154	0.193
10	**0.446**	0.338	0.382	0.137	0.158	0.145	0.188
12	**0.503**	0.373	0.415	0.138	0.164	0.148	0.187
15	**0.553**	0.437	0.453	0.141	0.174	0.156	0.187
18	**0.587**	0.498	0.485	0.144	0.183	0.168	0.187
20	**0.599**	0.523	0.497	0.149	0.188	0.172	0.187
22	**0.606**	0.537	0.502	0.160	0.174	0.177	0.187
25	**0.617**	0.560	0.513	0.195	0.160	0.185	0.188
50	**0.648**	0.622	0.531	0.523	0.142	0.250	0.187
100	**0.662**	0.646	0.540	0.571	0.270	0.307	0.185
250	**0.647**	0.641	0.529	0.604	0.574	0.353	0.184

V. RESULTS

Unless stated otherwise, results are for mature Berka accounts under Protocol I, pooled over five cohort seeds (300 accounts each) and 20 injection draws per account; paired tests use the primary seed. Every number below is generated from the repository's result tables.

A. Data-Efficiency Curves

The ridge-regularised Mahalanobis detector rises from 0.187 at $n = 1$ to 0.446 at $n = 10$ and 0.599 at $n = 20$, and gains less than 0.07 more: 0.648 at 50, 0.662 at 100 and 0.647 at 250 (Fig. 1 and Table II). Its relative-90 threshold is 20, 20, 22, 20, 22 across the five cohort seeds. Resampling the primary seed's accounts gives a 95% bootstrap interval of 18–20; that interval ignores cohort variation, and two of the five cohort draws give 22. The relative-95 threshold is 50. A saturating exponential describes the curve well ($R^2 = 0.955$, time constant 10.2 transactions) and fits better than a power law ($R^2 = 0.819$, $\Delta\text{AIC} = 16.7$ in favour of the exponential).

The detectors differ in rate more than in kind. The autoencoder saturates as early (relative-90 at 18, 18, 20, 22, 18) but

TABLE III

RELATIVE SUFFICIENCY THRESHOLDS: THE SMALLEST EVALUATED n REACHING AND KEEPING 90% (REL-90) OR 95% (REL-95) OF THE CURVE'S BEST OBSERVED PR-AUC, PER COHORT SEED (ONE VALUE WHEN ALL SEEDS AGREE). †: THE CURVE IS STILL RISING AT THE LARGEST n FOR AT LEAST ONE SEED, SO THE THRESHOLD IS OPTIMISTIC. —: NEVER REACHES AND KEEPS THE LEVEL WITHIN THE GRID. BEST: MEAN OVER SEEDS OF EACH SEED'S BEST PR-AUC.

	Rel-90	Rel-95	Best
Berka, mature accounts			
Mahalanobis, Protocol I	20, 20, 22, 20, 22	50	0.662
Protocol IC	18	20, 20, 25, 22, 22	0.633
Protocol P	22, 25, 50, 22, 25†	50, 50, 100, 50, 50	0.454
no timing features, I	10, 12, 12, 12, 12	15, 18, 15, 18, 15	0.529
no timing features, IC	10, 12, 10, 10, 10	12, 15, 12, 15, 12	0.632
no timing features, P	15, 18, 15, 18, 15†	20, 25, 25, 50, 25	0.726
Mahalanobis (Ledoit–Wolf)	50	50	0.646
Autoencoder	18, 18, 20, 22, 18	25, 25, 50, 50, 25	0.540
k-NN	100†	250, 250, 250, 250, 100	0.604
Isolation Forest	250†	250	0.353
Hybrid $\kappa = 25$, I	22, 25, 25, 25, 25	50	0.657
Hybrid $\kappa = 25$, IC	50†	50	0.599
Hybrid $\kappa = 25$, P	20, 25, 50, 22, 25†	50	0.410
Berka, early life ($n \leq 50$)			
Mahalanobis	15†	20	0.586
Autoencoder	18	—	0.507
Hybrid $\kappa = 25$	15†	20	0.583
Online Retail II			
Mahalanobis	250†	250	0.265
Mahalanobis (Ledoit–Wolf)	250	250, —, —	0.288
k-NN	—	—	0.314
Isolation Forest	250†	250, 250, —	0.242
Gaussian z	1	5	0.369
Hybrid $\kappa = 25$	250†	250	0.258
TabFormer (synthetic)			
Mahalanobis	250†	500	0.778
Mahalanobis (Ledoit–Wolf)	250†	500	0.711
k-NN	500†	500	0.731
Isolation Forest	100†	250	0.741
Hybrid $\kappa = 25$	250†	500	0.773

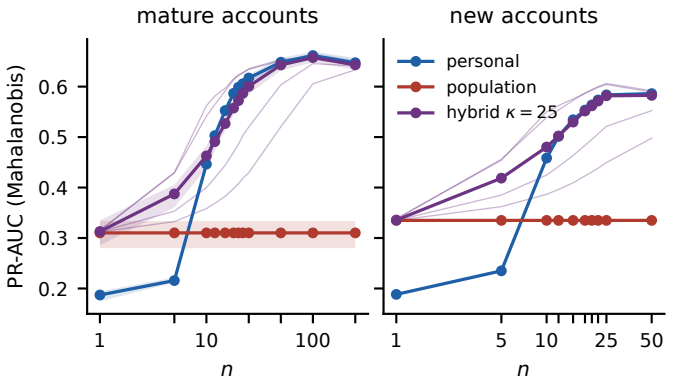


Fig. 2. Personal, population and shrinkage-hybrid Mahalanobis on Berka. Left: mature accounts (five seeds). Right: the early-life cohort (1,500 accounts, first 110 transactions). Faint lines: hybrids with $\kappa \in \{5, 10, 50, 100\}$. Bands: range over seeds.

TABLE IV

MAHALANOBIS ARMS ON MATURE BERKA ACCOUNTS. PERSONAL AND HYBRID ($\kappa = 25$) PR-AUC POOLED OVER FIVE COHORT SEEDS (POPULATION MODEL: 0.310). PAIRED DIFFERENCES ON THE PRIMARY SEED (300 ACCOUNTS; 95% BOOTSTRAP INTERVALS IN THE TEXT); HOLM-ADJUSTED WILCOXON p FOR HYBRID MINUS PERSONAL.

n	Pers.	Hyb.	Pers.−Pop.	Hyb.−Pers. (p)	Hyb.−Pop.
1	0.187	0.313	−0.159	+0.161 (<0.001)	+0.001
5	0.216	0.387	−0.122	+0.194 (<0.001)	+0.072
10	0.446	0.463	+0.114	+0.035 (0.002)	+0.149
15	0.553	0.527	+0.210	−0.002 (1.000)	+0.207
20	0.599	0.572	+0.267	−0.015 (1.000)	+0.252
25	0.617	0.601	+0.284	−0.006 (1.000)	+0.278
50	0.648	0.643	+0.310	+0.001 (0.707)	+0.311
100	0.662	0.658	+0.321	+0.001 (0.355)	+0.321
250	0.647	0.643	+0.309	−0.003 (0.392)	+0.306

at a lower ceiling (0.540 at $n = 100$). Ledoit–Wolf shrinkage, despite being designed for $n \leq d$, learns more slowly than the ridge estimate (0.338 against 0.446 at $n = 10$; relative-90 at 50). A likely reason, which we did not isolate, is the shrinkage target. Ledoit–Wolf pulls every variance toward the average variance, so a category the account has never used gets an ordinary variance. The small ridge instead leaves such directions with almost no variance, which makes them very expensive in the distance. k-NN does nothing useful below $n = 25$ (0.149 at 20) and reaches 0.604 at 250; LOF needs even more history. Isolation Forest never exceeds 0.353 (at $n = 250$) and is still rising there; k-NN is below it up to $n = 22$ and LOF from 25 to 100. Univariate z-scores on log amount are flat near 0.18 at every size: under Protocol I, an impostor's amount alone barely distinguishes it.

B. How Robust Is the Sufficiency Point?

Table III varies one thing at a time. Restricting injection to customer-initiated transactions (Protocol IC) moves the relative-90 threshold to 18; amount scaling (Protocol P) gives 22, 25, 50, 22, 25, with a much lower ceiling and a curve still rising at some seeds.

The calendar and gap features change the answer, and how they change it depends on the protocol. Under Protocol I the model without them saturates at 10, 12, 12, 12, 12 transactions but at a ceiling of 0.529 instead of 0.662. Under Protocol IC it loses nothing (best 0.632 against 0.633) and reaches 90% of its best at 10, 12, 10, 10, 10. Under Protocol P it is far better than the full model (best 0.726 against 0.454). The ranking quality that timing features add, and with it most of the gain between 10 and 20 transactions, is therefore specific to Protocol I. A plausible source, which we did not test, is the 32.2% of donor rows that the bank generates (interest, fees, sanctions): the bank posts them on fixed calendar days, and a substituted row carries one to an ordinary date. The defensible statement for Berka is that a personal Mahalanobis detector reaches 90% of its best ranking quality after 10 to 22 transactions: about 20 with calendar features under Protocol I, 10 to 12 without them. The metric matters as well: by ROC-AUC the same Protocol I curve reaches 90% of its best (0.889) at $n = 12$. A threshold is a property of a detector, a feature set, an anomaly definition, a metric and a dataset together, not of the person alone.

C. Personal, Population and Hybrid Models

The population model, fitted on 400 other accounts, scores 0.310 (0.280–0.334 across cohort seeds, which redraw both donors and evaluated accounts). The personal model is worse through $n = 5$ (difference -0.122, 95% interval [-0.139, -0.104]) and better from $n = 10$ (+0.114, [+0.091, +0.138]); the

crossover is at $n = 10$ at every seed. For k-NN it is at $n = 50$ (population k-NN scores 0.450) and for the autoencoder at $n = 12, 10, 10, 10, 10$.

The shrinkage hybrid starts exactly at the population model (difference +0.001, [-0.001, +0.003] at $n = 1$) and improves on the personal model below the crossover (+0.035 at $n = 10$, [+0.016, +0.053], Holm $p = 0.002$). Under Protocol I it then tracks the personal model. On the primary seed it differs from it by at most 0.015 from $n = 12$; the largest shortfall, -0.015 at $n = 18$ and 20, has a bootstrap interval just below zero ([-0.028, -0.002] at $n = 20$), and the Wilcoxon test, which tests a shift in the paired distribution rather than its mean, gives Holm $p = 1.000$. Pooled over the five seeds the largest shortfall is -0.029 at $n = 18$. Its relative-90 threshold is 22, 25, 25, 25, 25, later than the personal model's because it sits slightly below it between $n = 15$ and 22.

This favourable picture is specific to Protocol I and PR-AUC. Under Protocol IC the $\kappa = 25$ hybrid trails the personal model by about 0.1 at $n = 25$ (0.513 against 0.613) and reaches 90% of its best only at 50; under Protocol P it trails from $n = 15$ on and still at $n = 250$ (0.410 against 0.452); and by ROC-AUC under Protocol I it trails at $n = 20$ (0.802 against 0.865). Two differences contribute, and we did not fully separate them. One is the prior strength: $\kappa = 5$ recovers most of the Protocol IC gap (0.583 at $n = 25$). The other is standardisation (Section IV): the hybrid is not the personal model even as $\kappa \to 0$, and under Protocol P even $\kappa = 5$ trails at $n = 250$ (0.429). Table VIII varies the prior strength under Protocol I. Every κ from 5 to 100 starts at the population level and stays above it, and the personal data take over at a rate set by κ: with $\kappa = 5$ or 10 the hybrid exceeds the pre-specified $\kappa = 25$ at every n up to 25 and meets the personal curve by $n = 25$, whereas $\kappa = 100$ is still below it at $n = 50$. The pre-specified value is therefore not the best one on Berka; we report it as primary because it was fixed before the results, and a κ chosen on donor accounts would be a legitimate, untested improvement. The conclusion that survives every protocol is narrower than "use a hybrid": shrinking toward the population removes the early penalty, and after the crossover the personal model is at least as good. Switching from the hybrid to the personal model once n passes the crossover would therefore be the safer rule; we did not evaluate it as a separate arm.

a) Early life: The mature cohort learns from the history just before a late window, and that history may be easier to model than an account's first months. The early-life cohort (1,500 accounts, one seed; Fig. 2, right) evaluates rows 51–110 of each account from its first $n \leq 50$ transactions. Personal Mahalanobis reaches 0.458 at $n = 10$, 0.584 at 25 and 0.586 at 50, a lower ceiling than for mature accounts, and its relative-90 threshold is 15 transactions (measured against the best value within $n \leq 50$, since the curve is still rising slightly at 50). The population model scores 0.335, and the crossover is again at $n = 10$. The hybrid exceeds the personal model by +0.184 at $n = 5$ and is indistinguishable from it from $n = 12$ under Protocol I (all Holm $p > 0.8$). One result does not transfer: population k-NN (0.478) beats personal k-NN at every size up to 50 (0.459 at $n = 50$), so for a neighbour-based detector an account's first 50 transactions are not enough.

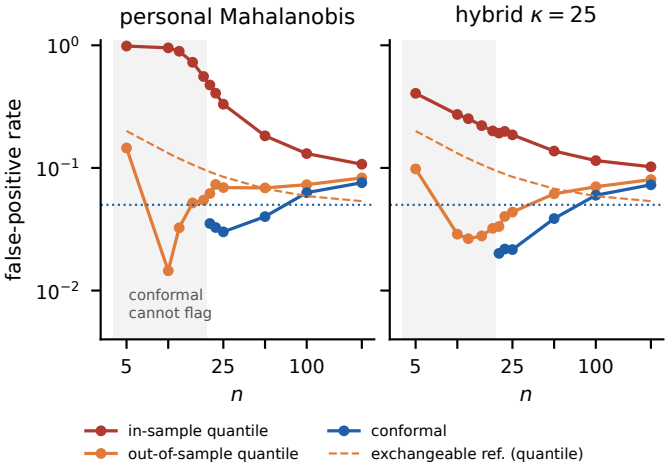


Fig. 3. False-positive rate of three threshold rules on mature Berka accounts (log scale). Dashed: the exchangeable expectation of the out-of-sample quantile rule, Equation (2). Dotted: $\alpha = 0.05$. Grey: $n < 19$, where no conformal threshold exists.

TABLE V
FALSE-POSITIVE RATE (FPR) OF THE THREE THRESHOLD RULES ON MATURE BERKA ACCOUNTS (PROTOCOL I, FIVE SEEDS, 20 DRAWS; NOMINAL $\alpha = 0.05$). "REF." IS THE EXCHANGEABLE EXPECTATION OF THE OUT-OF-SAMPLE QUANTILE RULE, EQUATION (2). RECALL, POOLED PRECISION AND THE SHARE OF ACCOUNTS WITH ANY ALERT ARE FOR THE CONFORMAL RULE, WHICH CANNOT FLAG BELOW $n = 19$.

	FPR, quantile rules			Conformal			
n	In-sample	OOS	Ref.	FPR	Recall	Prec.	Alerting
Personal Mahalanobis							
5	0.986	0.145	0.200	0.000	0.000	—	0.00
10	0.953	0.015	0.132	0.000	0.000	—	0.00
15	0.727	0.052	0.106	0.000	0.000	—	0.00
18	0.556	0.055	0.097	0.000	0.000	—	0.00
20	0.475	0.062	0.093	0.035	0.320	0.502	0.90
25	0.331	0.069	0.085	0.030	0.317	0.539	0.92
50	0.183	0.069	0.068	0.040	0.470	0.566	1.00
100	0.131	0.073	0.059	0.063	0.614	0.521	1.00
250	0.107	0.083	0.054	0.076	0.628	0.479	1.00
Hybrid Mahalanobis							
5	0.406	0.098	0.200	0.000	0.000	—	0.00
10	0.273	0.029	0.132	0.000	0.000	—	0.00
15	0.221	0.028	0.106	0.000	0.000	—	0.00
18	0.200	0.032	0.097	0.000	0.000	—	0.00
20	0.192	0.033	0.093	0.020	0.343	0.655	0.96
25	0.186	0.044	0.085	0.022	0.378	0.660	0.96
50	0.137	0.061	0.068	0.039	0.513	0.596	1.00
100	0.115	0.070	0.059	0.060	0.604	0.529	1.00
250	0.102	0.080	0.054	0.073	0.620	0.486	1.00

D. From Ranking to an Alert Threshold

Ranking quality does not give an alert threshold. Calibrating on the model's own training scores flags 0.953 of genuine transactions at $n = 10$ and still 0.183 at $n = 50$ (Fig. 3 and Table V), because a model scores the points it was fitted to as unusually typical. Out-of-sample scores remove most of this bias, but the 95th-percentile rule cannot be judged against 0.05: with exchangeable scores its expected FPR is 0.132, 0.085 and 0.068 at $n = 10$, 25 and 50 (Equation (2)). Measured against that reference the out-of-sample rule is conservative at small n (0.015 at $n = 10$) and close to exchangeable from $n = 25$ (0.069 against 0.085).

TABLE VI
WHAT THE ACROSS-ENTITY SPREAD OF PERSONAL MAHALANOBIS PR-AUC IS MADE OF (PRIMARY SEED). ICC(1): SHARE OF SINGLE-DRAW VARIANCE THAT IS BETWEEN ENTITIES. SD COLUMNS: SPREAD ACROSS ENTITIES IN ONE INJECTION DRAW, INJECTION NOISE WITHIN AN ENTITY, AND ESTIMATED TRUE BETWEEN-ENTITY SPREAD. LAST COLUMN: RELIABILITY OF THE 20-DRAW MEAN.

	n	ICC	SD one draw	SD noise	SD true	Rel. (20)
Berka	10	0.57	0.217	0.141	0.162	0.96
	25	0.40	0.184	0.144	0.119	0.93
	50	0.39	0.174	0.141	0.113	0.93
	250	0.38	0.177	0.143	0.112	0.92
Berka early	10	0.54	0.212	0.145	0.158	0.96
	25	0.51	0.210	0.145	0.147	0.95
	50	0.50	0.217	0.149	0.150	0.95
Retail II	10	0.38	0.119	0.095	0.074	0.93
	25	0.48	0.123	0.090	0.086	0.95
	50	0.45	0.127	0.094	0.085	0.94
	250	0.39	0.156	0.126	0.100	0.93
TabFormer	10	0.62	0.172	0.112	0.142	0.97
	25	0.59	0.199	0.127	0.154	0.97
	50	0.63	0.206	0.124	0.161	0.97
	250	0.71	0.198	0.107	0.167	0.98
	500	0.68	0.177	0.104	0.152	0.98

The conformal rule makes the guarantee explicit and exposes its cost. It cannot alert at all below $n = 19$. From $n = 20$ to 50 it keeps the FPR below 0.05 (0.030 at $n = 25$ and 0.040 at 50) with recall 0.317 and 0.470 and pooled precision 0.539 and 0.566. At $n = 100$ and 250 it exceeds 0.05 (0.063 and 0.076). The guarantee assumes the calibration scores are exchangeable with the test scores. The median history spans 98 days at $n = 25$, 226 at $n = 50$, 457 at $n = 100$ and 1,198 at $n = 250$, and the excess FPR at the two largest sizes is consistent with an account's older behaviour differing from its present. The hybrid behaves the same way with a lower FPR (0.022 at $n = 25$) and higher recall (0.378). The population model, calibrated by split conformal on held-out donor transactions, has FPR 0.077 with recall 0.119, above α (and 0.165 on TabFormer): other people's transactions are not exchangeable with a given account's either. Table VII gives every detector.

For an operator this replaces one number with two. On Berka a detector *ranks* nearly as well as it will from 10 to 22 transactions, depending on its features; it can carry a *valid* alert threshold only from 19 calibration points at $\alpha = 0.05$, more generally $1/\alpha - 1$; and on long histories the guarantee is exceeded again, consistent with older behaviour differing from the present.

E. What Differs Between Individuals

a) Most of the apparent spread is measurement noise, but not all: With one injection draw, per-account PR-AUC varies across accounts with standard deviation 0.184 at $n = 25$. Re-injecting the same account gives a within-account standard deviation of 0.144, so the intraclass correlation of a single-draw score is only 0.40 (Table VI and Fig. 4): at this size a single draw is more noise than signal. At $n = 10$ the ICC is 0.57, and on the other datasets 0.4–0.7, so there a single draw is roughly half noise. The true between-account standard deviation is 0.119 at $n = 25$ and 0.112 at $n = 250$. Individual differences in detectability are therefore real, about two-thirds

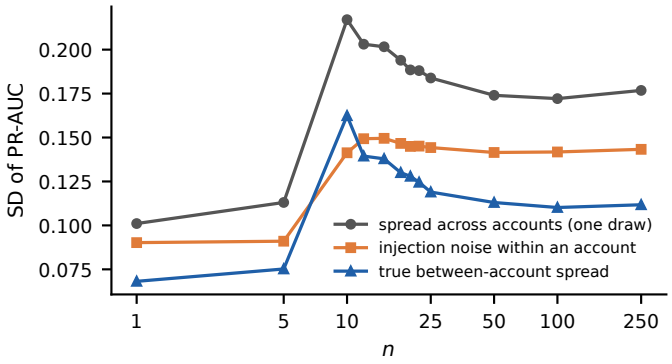


Fig. 4. Decomposition of the across-account spread of personal Mahalanobis PR-AUC on Berka (primary seed): the spread seen in one injection draw, the injection noise within an account, and the estimated true between-account spread.

the size of the raw spread, and they barely shrink between $n = 25$ and 250. The 20-draw mean has reliability 0.93, which is what the analyses below use.

b) Individual sufficiency points: On draw-averaged curves, 75% of accounts reach and keep 90% of their own best within 250 transactions. The other 25% end, at $n = 250$, more than 10% below their own best: for them the oldest history hurts. Among the accounts with a threshold the median is 50 transactions (interquartile range 18–100), above the 20, 20, 22, 20, 22 read off the mean curve. Two explanations fit: averaging smooths individual curves that rise at different points, and the reach-and-keep rule applied to noisier per-account curves pushes thresholds up. The split-half reliability of an account's threshold is 0.77, so the thresholds are consistent across draws, which does not exclude a shared upward bias. In the early-life cohort the per-account median is 20 (interquartile range 15–50), close to the mean curve.

c) Observed variability does not predict them on Berka: We correlated each account's threshold with eight descriptors of its last 250 transactions (amount coefficient of variation, robust coefficient of variation, log-amount interquartile range, operation and `k_symbol` entropy, operation diversity, inter-arrival coefficient of variation and credit share). All $|\rho| \leq 0.09$ and all Holm-adjusted $p = 1$ over 225 accounts. With 80% power the test detects an observed $|\rho|$ of 0.19, a true correlation of 0.21 after correcting for the threshold's reliability, so on Berka a moderate dependence of the requirement on these descriptors is ruled out and a weak one is not. The same descriptors do predict the *ceiling*: accounts with higher operation entropy reach higher PR-AUC ($\rho = +0.25$) and accounts with more variable amounts lower ($\rho = -0.23$), plausibly because a distinctive profile is easier to separate from an impostor's.

d) Variability under control: the simulation: The simulation generates 240 individuals in six archetypes whose behavioural parameters (amount dispersion, number of merchants and locations, timing regularity, drift) are set by design, with six synthetic anomaly classes (amount, merchant, category, location, timing and a multivariate combination) injected into a 100-transaction window (Fig. 5). Here the requirement differs

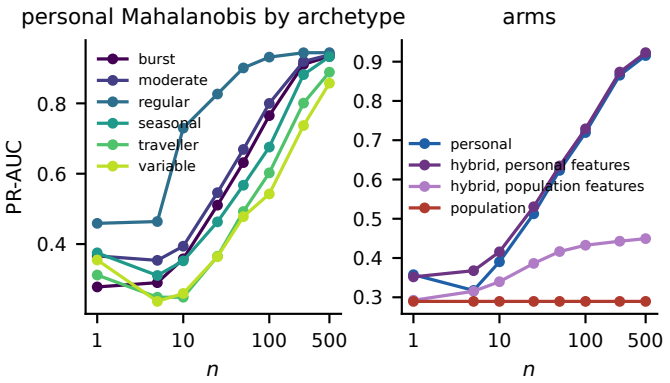


Fig. 5. Simulation (240 individuals in six archetypes, five seeds, 20 injection draws). Left: personal Mahalanobis by archetype. Right: personal, population and the two hybrid variants.

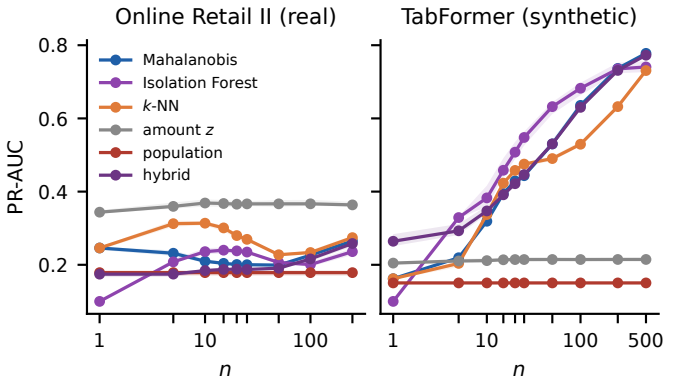


Fig. 6. Personal detectors, the population Mahalanobis model and the $\kappa = 25$ hybrid on Online Retail II (left) and TabFormer (right), Protocol I, mean over three cohort seeds (bands: range over seeds).

tenfold between archetypes: relative-90 is 50 for the regular archetype, 250 for moderate, seasonal and burst behaviour, 250, 500, 500, 250, 250 for travellers and 500 for the most variable, while the best values are similar (0.857–0.944). Across all 240 individuals the generating parameters predict an individual's threshold strongly (number of merchants $\rho = +0.73$, amount dispersion +0.68, locations +0.68), and so do the observed descriptors (merchant entropy +0.72, category entropy +0.74); the thresholds themselves are measured with split-half reliability 0.99.

These correlations are between archetypes. Within an archetype every generating parameter is identical, so there is no designed variability for the descriptors to find, and they do not predict the threshold (pooled within-archetype $|\rho| \leq 0.15$, all Holm $p \geq 0.10$; the largest single-archetype value is $|\rho| = 0.43$). The simulation therefore shows that a difference in behavioural regime can change the requirement roughly tenfold (the most variable archetype's value is the largest size evaluated, so it is censored). It cannot show whether observed descriptors would detect differences between individuals within a regime, because by design it contains none.

Two further simulation results bear on earlier sections. First, a hybrid variant that computes novelty features against the population's merchant list rather than the person's, as an earlier version of this study did, reaches only 0.449 at $n = 500$. The same hybrid with person-referenced novelty features reaches 0.923, matching the personal model (0.916) and exceeding it at small n (0.416 against 0.390 at 10). The hybrid failure reported in that earlier version came from this choice, not from shrinkage. Second, the conformal rule exceeds α badly on short simulated histories and approaches it on long ones: FPR 0.415 at $n = 25$, 0.138 at 100 and 0.052 at 500. Simulated individuals drift and change with the season by construction, and a short history leaves many of a person's merchants unseen, so window transactions are flagged as novel more often than history transactions are. Either would break exchangeability; we did not separate them.

e) Recency, span and drift: Holding the count at $n = 50$, taking every fourth transaction instead of the 50 most recent widens the median span from 226 to 946 days and changes Mahalanobis PR-AUC by -0.033 (Holm $p < 0.001$); every second transaction changes it by -0.010 ($p = 0.002$). A contiguous block of 50 that ends 50, 100 or 150 transactions before the window keeps density fixed and changes only age: -0.029, -0.053 and -0.070 ($p < 0.001$, $p < 0.001$ and $p < 0.001$; the oldest block ends a median 731 days before the window). For k-NN the corresponding changes are -0.103 (stride 4) and -0.043 (lag 150).

Behaviour does drift. Measured in the coordinates of an account's most recent 100 transactions, the Mahalanobis distance to its transactions 101–250 back has median 0.69. A sweep comparing each earlier 25-transaction block with the most recent 25 rises from 0 (by definition) to 1.75 at lag 25 and peaks at 3.63 at lag 175 (3.25 at lag 200). We did not estimate a same-period noise floor, so much of the step from lag 0 to 25 may be sampling noise in a 25-row covariance; the further rise with lag is the evidence of drift. An account's drift does not predict its own PR-AUC change from $n = 100$ to 250 ($\rho = -0.089$, $p = 0.125$); the gain itself has reliability 0.93, and the attenuation-corrected correlation is -0.09 (95% interval [-0.200, +0.022]). Drift is real, accumulates with lag and is consistent with the conformal excess at long histories, but it does not explain which accounts gain or lose from older data.

F. Other Datasets

a) Online Retail II: Retail customers do not behave like Berka accounts, and the sufficiency question gets a different answer (Fig. 6 and Table III). The best detector is the simplest: a z-score on the line amount reaches 0.369 and has its relative-90 threshold at 1 line, plausibly because customers differ mainly in the scale of what they buy and one previous line already shows it. Every multivariate detector is worse at every size. Personal Mahalanobis falls from 0.246 at $n = 1$ to 0.199 at $n = 50$ before recovering to 0.265 at 250, and k-NN, LOF and the autoencoder peak at $n = 5$ or 10 (best k-NN 0.314) and then decline. The reason is that a line is not an independent observation. Lines arrive in invoices, and a 25-line history spans a median of 2 invoices (10 at $n = 250$; the 60-line window spans 3). A multivariate model of 25 lines is in effect

a model of two baskets. We take this as the likely reason the multivariate detectors fit the baskets rather than the customer, but did not test it directly; removing the timing features does not change the pattern (0.270 at $n = 1$, 0.218 at 50).

Calibration fails too. The conformal Mahalanobis threshold flags 0.373 of genuine lines at $n = 25$, consistent with the window's baskets not being exchangeable with the history's; on the amount alone, where baskets matter less, the conformal FPR is 0.051. Population models are weak (best 0.219, robust_z, against a base rate of 0.10) and lose to the personal model from $n = 1$, and the hybrid, which starts at the population model, pays for it: its difference from the personal model is -0.045 at $n = 5$ (Holm $p < 0.001$) and -0.005 at 250. Shrinking toward the population helps only when the population model beats the personal one at small n, which can be checked on donor accounts before deployment.

b) TabFormer: The synthetic card data have the merchant, category and location fields Berka lacks and histories up to 500 transactions, and they give a third answer. Personal Mahalanobis is still rising at $n = 500$ (0.530 at 50, 0.636 at 100, 0.778 at 500), so the relative-90 threshold of 250 is relative to an unconverged best and there is no sufficiency point within the grid. The detector ranking reverses relative to Berka: Isolation Forest, the weakest multivariate detector on bank data, is the best from $n = 5$ to 100 (0.383 at 10, 0.632 at 50, 0.682 at 100) and reaches 90% of its best at 100; Mahalanobis catches up at 250 and leads at 500. Population models are weak (0.150; best 0.177) and lose to the personal model from $n = 1$. The hybrid still helps at small n (+0.124 over personal at $n = 1$, +0.044 at 10), unlike on Retail. The hybrid differs from the population arm by including the personal novelty features (merchant, category or state seen before), and it already scores 0.264 at $n = 1$ against the population model's 0.150, consistent with those features carrying the early gain on TabFormer but not on Retail (hybrid 0.174 at $n = 1$). At 500 the difference is -0.006.

Calibration fails in the opposite direction to Retail. At $n = 25$ the in-sample rule flags 0.809 of genuine transactions and the conformal rule only 0.003, with recall 0.009: the rule is valid but useless. A plausible cause, which we did not isolate, is that leaving out the only transaction at a merchant makes that transaction look novel, so out-of-sample scores on short card histories are inflated. By $n = 100$ the conformal rule is near nominal (0.046, recall 0.438), and at $n = 500$ it slightly exceeds α (0.060), as on long Berka histories. Under Protocol P every multivariate detector stays near the base rate (Mahalanobis 0.189 at 500) and only the amount z-score detects scaled amounts (0.415), whereas on Berka personal Mahalanobis does detect them; we did not investigate the difference.

G. Disparate-Impact Audit

Sparse-history users are the people this study is about, so we test whether false positives concentrate by district economics, joining each account to its district's average salary, unemployment rate, urbanisation and region [44]. For personal and hybrid Mahalanobis under the conformal rule at $n = 25$, 50 and 100 we test Spearman correlations of an account's draw-averaged FPR with the three continuous variables (Holm within each set of three) and a Kruskal–Wallis test across eight regions. One of the 18 correlations is significant within its family: for the hybrid at $n = 100$, higher district salary goes with lower FPR ($\rho = -0.15$, Holm $p = 0.027$; uncorrected $p = 0.009$), which does not survive correction across all 18 tests. No region effect is significant. The audit is ecological [52]: Berka has no individual demographic field, so it cannot show that individuals are treated alike.

VI. Discussion

a) What the evidence establishes: On real bank histories a personal Mahalanobis detector ranks impostor transactions nearly as well after 10 to 22 of the person's own transactions as it ever will. The value is stable across cohort draws for a fixed configuration, but it moves with the feature set, the anomaly definition, the metric, the detector and the dataset (Sections V-A, V-B and V-F). It is not a constant of human behaviour; it is the sample size at which one estimator of one representation stops improving on one kind of anomaly. Below about ten transactions a population model is better on bank data, and shrinking toward it removes that penalty under every anomaly definition we tested; after the crossover the personal model is at least as good (Section V-C). A detector that ranks well is not yet one that can alert: a valid threshold needs its own minimum history and is exceeded again on long histories (Section V-D). People differ genuinely in how well they can be modelled, and observed variability predicts that, but on bank data it does not predict who needs more history (Section V-E).

b) What the other datasets change: The bank result does not generalise as a number (Section V-F). On retail purchase lines one feature that captures a customer's purchase scale is learned from a single line, and the multivariate detectors learn slowly or not at all, plausibly because lines arrive in invoices and 25 lines cover about two baskets. On synthetic card data, which carry merchant and location fields and long histories, the best detector is still improving at 500 transactions, and Isolation Forest, among the weakest detectors on bank data, is the best between 5 and 100 transactions. Two regularities hold on every dataset: the in-sample threshold flags most genuine transactions at small n, and the population model beats the personal one early only where it is stronger at $n = 1$, which can be checked on donor accounts. The practical point follows: a system must measure its sufficiency point on its own data, detector and features.

c) What remains interpretation: Several explanations in this paper are plausible and untested: that Ledoit–Wolf learns more slowly than the ridge estimate because of its shrinkage target; that the timing features' contribution under Protocol I comes from bank-generated rows posted on fixed calendar days; that the conformal excess at long histories reflects behaviour the person has since changed, which the drift measurement is consistent with but does not isolate; that Retail's poor multivariate learning and failed calibration come from invoice structure; that customers there differ mainly in purchase scale; that novelty features carry the hybrid's early gain on TabFormer; and that leave-one-out novelty inflates

out-of-sample scores on short card histories. The last two point in different directions from the simulation, where window transactions look more novel than history transactions; we did not reconcile them. Whether observed descriptors could predict individual requirements on data with richer channels is open: the simulation contains no within-regime variability by design, so it cannot answer it.

d) Practical guidance: For a system resembling Berka, the evidence supports five decisions. Start a new account on the population model or a shrinkage hybrid, after checking on donor accounts that the population model beats a one-transaction personal model, and switch to the personal model once the history passes the crossover; keeping a fixed-strength hybrid afterwards can cost accuracy. Measure the sufficiency point for the features actually used: on Berka it is about 20 transactions with calendar features and about 10 without, and neighbour- and tree-based detectors need several times more. Set alert thresholds out of sample; an in-sample threshold on a short history is not a threshold. Do not claim a nominal false-positive rate before $1/\alpha - 1$ calibration points exist, and do not assume that a longer calibration history is a better one. Finally, report per-person performance only from repeated evaluation, because a single injection draw is at best about half signal.

VII. Limitations

a) Data: The real banking evidence is one dataset from one Czech bank, 1993–1998, with day-resolution timestamps and no merchant, category or location fields. Online Retail II is real but records purchase lines of a UK online gift retailer, many of whose customers are businesses, not payments by individuals; TabFormer, which has the merchant and location channels Berka lacks, is synthetic. Real history sizes stop at 250 because few accounts are deeper. The mature cohort requires at least 310 transactions, which selects the more active accounts; the early-life cohort covers the first months of an account but still requires it to reach 110 transactions.

b) Protocol: An impostor transaction is a transaction the person did not make, not fraud. Neither protocol models an adversary who imitates the victim, which would be the hardest case. The 10% injection rate is far above realistic fraud prevalence, so absolute PR-AUC values and pooled precision do not transfer to deployment. Whether the curves' shape transfers is untested; by ROC-AUC, which does not depend on prevalence, the Protocol I relative-90 threshold is 12 rather than 20. The false-positive rates on genuine transactions do not depend on prevalence. History count and span cannot be separated at fixed count, which the stride and lag experiments address only partly.

c) Modelling: The hybrid's prior strength was fixed before the results and is not the best value on Berka; choosing it on donor accounts, and the switch-to-personal rule we recommend, are untested. The hybrid also differs from the personal model in standardisation, which confounds its later shortfall. The autoencoder is small, and Isolation Forest and LOF use library defaults. Relative thresholds are read from a discrete grid, so a threshold of 20 means "between 18 and 20". The personal conformal rule calibrates on leave-one-out or cross-fitted scores of the training history rather than on a separate split, so its guarantee holds only approximately: each calibration score comes from a model fitted on slightly less data than the one that scores the window. A split would be exact but would halve an already short history.

d) Statistics: Berka mature results use five cohort draws; the early-life cohort uses one, Retail and TabFormer three, and the simulation five. The account bootstrap of the sufficiency threshold uses the primary seed and does not include cohort variation. With 300 paired accounts, Wilcoxon p values are tiny for effects of no practical size, so we report effect sizes and intervals alongside them. The variability analysis has 80% power only for true correlations of about 0.21 or larger.

VIII. Ethical Considerations

a) Deviation is not suspicion: Our detectors flag transactions that differ from a person's past. In life those are holidays, house moves, medical bills, a new job, a bereavement, leaving an abusive relationship. A system that treats deviation as suspicion penalises people for changing.

b) Errors are not symmetric: A missed anomaly costs an institution money; a false positive can mean a declined card abroad or a frozen account in an emergency. At $\alpha = 0.05$ a person who makes 40 transactions a month should expect about two false alerts a month on average even when the guarantee holds, and the guarantee does not hold below 19 calibration transactions, on long bank histories, on Retail or in the simulation (Sections V-D to V-F).

c) Sparse-history users: The people with the least history are new customers, young people, recent immigrants and people returning to formal banking. On Berka a personal model is worse than a population model for them, and on every dataset an in-sample threshold would flag most of what they do. An out-of-sample threshold protects this group; a population or hybrid model helps only where, as on Berka, the population model is informative, and even its calibrated false-positive rate exceeded the nominal level.

d) Discrimination: Behavioural features are proxies for life circumstances: irregular income, cash use and mobility correlate with class, occupation, immigration status and disability. Our district-level audit (Section V-G) finds no relationship that survives correction across all tests, but it is ecological and covers 300 accounts. It narrows the concern and does not clear it; deployment would need an individual-level audit on the axes Berka cannot provide.

e) Privacy and data handling: Per-person modelling requires retaining personal transaction histories. On Berka 10 to 22 recent transactions carry most of the ranking quality, a fixed number of older transactions is worse than the most recent ones, and long histories exceed the calibration guarantee, so shorter retention appears to cost little there, a case where privacy and performance point the same way. All three datasets are public; we redistribute no raw records and make no attempt at re-identification.

f) Appropriate use: These thresholds describe when a personal baseline ranks well on these data. They are not a licence to deploy a detector after N transactions, and a flagged transaction is not grounds for adverse action against a person.

IX. Conclusion

We measured how much of a person's own transaction history an unsupervised anomaly detector needs, with paired curves, provenance labels, three threshold rules judged against their own references, and repeated injection. On real bank accounts the answer is 10 to 22 transactions for ranking, depending mainly on whether calendar features are used, a population model before about ten, and at least 19 out-of-sample calibration points before a false-positive guarantee means anything. On long histories that guarantee is exceeded again, consistent with behaviour that changes over time.

The number does not travel. On retail purchase lines one feature is learned at once and richer detectors barely learn; on synthetic card data with merchant and location fields the best detector is still improving at 500 transactions and the detector ranking changes. Even on one dataset the number moves with the anomaly definition and the metric. What does travel is the method: a system can measure its own sufficiency point on donor data with the same protocol, and should, before deciding when a new customer's own history is trusted.

People genuinely differ in how well they can be modelled, but on bank data observed behavioural variability does not predict who needs more history, so a per-person guarantee cannot be read off a population curve. Three extensions would sharpen these results: a real dataset with merchant fields and individual demographics, which would allow an individual-level fairness audit; choosing the hybrid's prior strength and a switch-to-personal rule on donor accounts; and a simulation with variability inside each behavioural regime, which would test whether observed descriptors can find individual requirements at all.

Appendix A
Additional Results

Table VII gives recall under the conformal rule for every personal detector and the hybrid on mature Berka accounts, and Table VIII the PR-AUC of the hybrid at every prior strength κ.

Appendix B
Reproducibility

a) Environment: Python 3.14.6 with pandas 3.0.6, NumPy 2.5.3, scikit-learn 1.9.1, SciPy 1.18.1, Matplotlib 3.11.2 and openpyxl 3.1.5 (exact versions in `requirements-lock.txt`). No GPU is used; the autoencoder is implemented in NumPy.

b) Data integrity: Every Berka table matches the record counts published by the dataset's authors and is byte-identical, after line-ending normalisation, across two independent mirrors (digests in the repository). TabFormer's 24,386,900 rows are counted by a full pass over the file; Online Retail II is read from the UCI archive and its SHA-256 is recorded.

c) Regression gate: Before any result is reported, `experiments/v2_regression.py` runs the pipeline with a single injection draw, in-sample thresholds and the feature construction of the repository's reference implementation (`src/experiment_engine.py`), and checks that PR-AUC, FPR and flag rate equal that implementation's committed outputs for the personal, population and $\kappa = 25$ hybrid arms. Over 40 accounts, seven history sizes and two protocols the largest difference is 2×10^{-16}. The same script checks the ranking metrics against scikit-learn and Equation (2) against Monte Carlo.

d) Seeds: Cohort seeds are 20260920 to 20260924. Per-entity injection draws are seeded from (cohort seed, entity identifier, draw, protocol); entity identifiers for TabFormer are hashed with CRC-32, which is stable across processes.

e) Parameters: Table IX lists the fixed parameters and Table X maps results to scripts; every figure is regenerated by `paper/make_figures.py` from the analysis tables.

Data and Code Availability

The code and results of the first version of this study are available at https://github.com/briyandyju09/personal-history-anomaly-detection. The revision-2 pipeline used for every number in this article, with its per-entity results, analysis tables and figure scripts, is available from the author on request. The Berka, Online Retail II and TabFormer data are public and are downloaded by the repository's loaders; no raw records are redistributed.

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TABLE VII
CONFORMAL-RULE RECALL (AND FPR) ON MATURE BERKA ACCOUNTS, FIVE SEEDS, 20 DRAWS. THE RULE CANNOT ALERT BELOW $n = 19$.

Detector	$n = 20$	$n = 25$	$n = 50$	$n = 100$	$n = 250$
Gaussian z	0.042 (0.004)	0.049 (0.004)	0.056 (0.004)	0.059 (0.006)	0.060 (0.006)
Robust z	0.028 (0.009)	0.033 (0.011)	0.037 (0.011)	0.040 (0.011)	0.039 (0.009)
Mahalanobis	0.320 (0.035)	0.317 (0.030)	0.470 (0.040)	0.614 (0.063)	0.628 (0.076)
Mahal. (LW)	0.244 (0.026)	0.298 (0.026)	0.493 (0.042)	0.606 (0.065)	0.624 (0.077)
k-NN	0.017 (0.020)	0.015 (0.012)	0.300 (0.033)	0.498 (0.061)	0.589 (0.075)
LOF	0.045 (0.027)	0.098 (0.119)	0.026 (0.037)	0.095 (0.017)	0.565 (0.079)
Isolation Forest	0.008 (0.006)	0.010 (0.006)	0.035 (0.010)	0.124 (0.032)	0.204 (0.045)
Autoencoder	0.184 (0.021)	0.202 (0.018)	0.342 (0.035)	0.442 (0.055)	0.462 (0.068)
Hybrid $\kappa = 25$	0.343 (0.020)	0.378 (0.022)	0.513 (0.039)	0.604 (0.060)	0.620 (0.073)

TABLE VIII
HYBRID PR-AUC BY PRIOR STRENGTH κ (MATURE BERKA, FIVE SEEDS).

	$n = 1$	$n = 5$	$n = 10$	$n = 25$	$n = 50$	$n = 250$
$\kappa = 5$	0.315	0.430	0.562	0.634	0.651	0.641
$\kappa = 10$	0.314	0.429	0.541	0.636	0.654	0.642
$\kappa = 25$	0.313	0.387	0.463	0.601	0.643	0.643
$\kappa = 50$	0.312	0.353	0.400	0.520	0.603	0.641
$\kappa = 100$	0.312	0.333	0.358	0.429	0.520	0.633
personal	0.187	0.216	0.446	0.617	0.648	0.647

TABLE IX
EXPERIMENTAL PARAMETERS.

Parameter	Value
Evaluation window	60 transactions (100 in the simulation)
Injected anomalies	6 per window (10%), 20 draws per entity and protocol
History sizes, Berka mature	1, 5, 10, 12, 15, 18, 20, 22, 25, 50, 100, 250
History sizes, Berka early life	1, 5, 10, 12, 15, 18, 20, 22, 25, 50
History sizes, Online Retail II	1, 5, 10, 15, 20, 25, 50, 100, 250
History sizes, TabFormer	1, 5, 10, 15, 20, 25, 50, 100, 250, 500
Entities	300 evaluated, 400 donors per cohort seed
Cohort seeds	5 (Berka mature), 3 (Online Retail II, TabFormer), 1 (early life), 5 (simulation)
Nominal FPR α	0.05
Hybrid κ	25 (primary); 5, 10, 50, 100
Out-of-sample scores	exact leave-one-out (distance detectors, hybrid); 10-fold cross-fitting (Isolation Forest, LOF, autoencoder)
Population calibration	split conformal on 2,000 held-out donor rows
Bootstrap	2,000 (intervals), 1,000 (thresholds)

TABLE X
RESULT-TO-SCRIPT MAP (`EXPERIMENTS/`).

Result	Script
Regression gate	`v2_regression`
Curves, arms, calibration, reliability (all datasets)	`v2_run`
Recency, span and drift	`v2_recency`
Simulation	`v2_simulation`
All tables and summary statistics	`v2_analyse`

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